Northern Ledger

VOL. 1, NO. 040

2026-03-30

FEATURES

Government indecision is still costing lives in the UK

kellybrendel · The BMJ Blog

The government must act now, or be faced with much tougher decisions and less popular choices as the winter kicks in, says John Middleton

One person in 55 is now infected with coronavirus in the UK . The odds are that every time you step onto a crowded bus or train carriage you meet someone who is infected with covid-19. Every time you are sitting in a restaurant maybe, or walking down a supermarket aisle, you will likely meet someone who is infected too. In early October one in 12 children were estimated to have had covid, and on 22 October there were 180 deaths in the UK. Every four hours, as many deaths are occurring in the UK as New Zealand has had for the whole pandemic.

We are still in groundhog day , with a few new repeating scenes. The secretary of state for health and social care, Sajid Javid, has repeated what he said when first taking up the role: that " there could be 100 000 cases a day ," yet he doesn't see this as a need to act, either then or now. English directors of public healt h have again broken from national guidance that they know is inadequate for their areas. NHS leaders and the BMA have called for urgent action "to protect the NHS."

Plan A of the government's winter plan has only ever been the "do as little as possible" option. The government's eggs are all in the vaccination basket: offer flu vaccination and a third dose of the covid vaccine as a booster (or not) then let everything else run loose and it's the people's fault if they get ill or infect others.

Malta, Spain, Portugal, and the Netherlands have now overtaken the UK and Israel in the vaccination stakes . France, which had a high level of vaccine hesitancy, has climbed up to a comparable level of coverage to the UK, mainly because the French are using vaccine passports. Eastern and southern Europe, along with the Baltic countries , are now experiencing severe epidemics. They have had high levels of vaccine hesitancy and lower vaccine uptake. In some cases, health systems have been less well prepared to implement mass vaccination, or lost out in the chaotic EU central purchase earlier this year.

Vaccination is still the best hope we have at reducing the virus to very low levels. Countries that have the highest rates of vaccination have seen an encouraging levelling of death rates throughout this year. Yet there is still no evidence of collective ("herd") immunity even in countries where there is a very high level of vaccination or where there have been persistent waves of infection, as in Iran. Reinfection occurs, especially with the delta variant , and vaccine efficacy declines over time . Vaccines still have a high level of efficacy against the delta variant, although less than with early variants. Yet where there are areas of low vaccine uptake within countries and with social mixing, these weaken a country's ability to suppress the virus. In the UK there are areas where public confidence in the vaccine is being improved with interventions from local public health teams and community support, but more investment is needed in this localised action. Vaccines prevent serious illness as measured by hospital admission and death but do they prevent debilitating illness—persistent or long covid ? Social mixing without masking enables the virus to find new unprotected, vulnerable people to infect and harm. Every time the virus finds a new host, new mutations occur, and new, potentially more vaccine resistant, variants will be formed in the UK's "new variant factory. " So vaccination is not the sole answer.

UK politicians have been too attracted to single technological fixes (testing, vaccination, the covid app, Nightingale hospitals), when what we need is the "Swiss cheese model" —using everything we've got at the same time to prevent viral spread. All of the European countries now enjoying high levels of double vaccine coverage with MRNA vaccines have still kept their social measures in place, in differing degrees throughout the summer. Masks have continued to be a requirement in public enclosed spaces in France, Portugal, and Spain. Germany requires FFP2 masks in enclosed public spaces and vaccine passports. Gatherings in public places have been strictly limited in Portugal and Spain until recently, despite high levels of vaccine coverage. France, Ireland, Montenegro, and Israel are other examples of countries requiring vaccine passports.

Living with covid does mean masking up in enclosed spaces, meeting outdoors when possible, working from home if we can, and if we can't do without our night time entertainment it means vaccine passports. Vaccination, or evidence of a recent negative test, are a small price to pay for "normality." Every vaccinated person has their vaccination card, and at least 10 million people now have the NHS app. It would seem a very small step to have to show it to gain access to somewhere you want to go—especially where employers should be protecting their staff. Millions of airline passengers have accepted the testing regimes of countries they wish to go to and the mask requirement on planes. The requirements in the UK plan B need to be put back into law for consistent control of the virus across the UK. None of them are major restrictions on our freedom.

The government has now asked local authorities what they think about moving towards plan B . They are, once again, caught stalling on actions that experts called for in the summer. Government ministers trivialise what is at stake with the ill informed debate about masking in Parliament, with careless contributions from the leader of the House of Commons , the care minister , and now the chancellor . They have chosen to pick a destructive and meaningless fight over the non-issue of face to face consultations in primary care, when they should have been praising the extraordinary effort of healthcare staff who've rolled out five rounds of covid and flu vaccinations. High levels of satisfaction with GP services continue , patients who need to get a face to face consultation get them, and the digital revolution envisaged by the NHS Long Term Plan has arrived.

The Download: reawakening frozen brains, and the AI Hype Index returns

Thomas Macaulay · MIT Technology Review

This is today's edition of The Download , our weekday newsletter that provides a daily dose of what's going on in the world of technology.

This scientist rewarmed and studied pieces of his friend's cryopreserved brain

L. Stephen Coles's brain sits in a vat at a storage facility in Arizona. It has been held there at a temperature of around −146 degrees °C for over a decade, largely undisturbed. Before he died in 2014, Coles had the brain frozen with an ambitious goal in mind: reanimation.

His friend, cryobiologist Greg Fahy, believes it could be revived one day. But other experts are less optimistic.

Still, Fahy's research could lead to new ways to study the brain. And using cryopreservation for organ transplantation is becoming a viable reality.

Read the full story to find out what the future holds for the technology .

—Jessica Hamzelou

The AI Hype Index

Separating AI reality from hyped-up fiction isn't always easy. That's why we've created the AI Hype Index—a simple, at-a-glance summary of everything you need to know about the state of the industry. Take a look at this month's edition .

MIT Technology Review Narrated: how Pokémon Go is giving delivery robots an inch-perfect view of the world

Pokémon Go was the world's first augmented-reality megahit. Released in 2016 by Niantic, the AR twist on the juggernaut Pokémon franchise fast became a global phenomenon. "500 million people installed that app in 60 days," says Brian McClendon, CTO at Niantic Spatial, an AI company that Niantic spun out last year.

Now Niantic Spatial is using that vast trove of crowdsourced data to build a kind of world model —a buzzy new technology that grounds the smarts of LLMs in real environments. The firm wants to use it to help robots navigate more precisely.

—Will Douglas Heaven

This is our latest story to be turned into an MIT Technology Review Narrated podcast, which we're publishing each week on Spotify and Apple Podcasts . Just navigate to MIT Technology Review Narrated on either platform, and follow us to get all our new content as it's released.

The next era of space exploration

Our footprint in the solar system is rapidly expanding. Programs to build permanent Moon bases and find life on Mars have transitioned from science fiction to active space agency missions. The scientists behind them will not only shed new light on the cosmos, but also reveal where humanity is headed.

To examine what the future holds in store, MIT Technology Review features editor Amanda Silverman will sit down today with award-winning science journalist and author Robin George Andrews for an exclusive subscriber-only Roundtable conversation about "The Next Era of Space Exploration." Register here to join the session at 16:00 GMT / 12:00 PM ET / 9:00 AM PT.

The must-reads

I've combed the internet to find you today's most fun/important/scary/fascinating stories about technology.

1 OpenAI is shutting down AI video generator Sora The app attracted at least as much controversy as acclaim. (CNBC)

1. Closing it means saying goodbye to $1 billion from Disney. (BBC)
2. OpenAI is cutting back on side projects ahead of an expected IPO. (WSJ $)
3. But it's focusing its efforts on building a fully automated researcher. (MIT Technology Review)

2 A judge suspects the Pentagon is illegally punishing Anthropic She labelled the DoD's ban "troubling." (Bloomberg)

1. Anthropic and the Pentagon are facing off in court. (Guardian)
2. The DoD wants AI companies to train on classified data. (MIT Technology Review)

3 Meta has been ordered to pay $375 million for endangering children online Prosecutors said the company knew it put children at risk. (Engadget)

1. Meta is offering its top talent stock options as incentives for its AI push. (CNBC)

4 Arm will sell its own computer chips for the first time It's aimed at data centers that run AI tasks. (NYT $)

1. Arm stock jumped 13% on the news. (CNBC)

5 Manus's founders have been barred from leaving China following Meta's takeover Beijing is reviewing the $2 billion acquisition of the AI startup. (FT $)

6 Baltimore has sued xAI over Grok's fake nude images The chatbot allegedly violated consumer protections. (Guardian)

1. There's a big market for pornographic deepfakes of real women. (MIT Technology Review)

7 NASA plans to send a nuclear-powered spacecraft to Mars in 2028 It'll take a payload of Ingenuity-class helicopters to the Red Planet. (NYT $)

1. NASA also wants to put a $20 billion base on the Moon.

Academic medicine and publishing from developing countries

kellybrendel · The BMJ Blog

Samiran Nundy, Atul Kakar, and Zulfi Bhutta have published a book titled How to Practice Academic Medicine and Publish from Developing Countries? A Practical Guide . It's a book that will be extremely useful to the growing number of academics working in low and middle income countries. The book is published by Springer and will from 30 October be available open access, meaning you can access it for free as often as you want. Hard copies will also be available for a fee. I felt privileged to be asked to write the foreword, and what follows is an edited version of my foreword.

When I became an assistant editor at The BMJ in 1979 and began to process scientific papers submitted to the journal it was extremely unusual to see, let alone publish, a paper from a low income country. I remember being surprised by high quality papers coming from Bangladesh, a country that Henry Kissinger called a "basket case." Those papers came from the International Centre for Diarrhoeal Disease Research, Bangladesh, which I learnt later was a creature of the Cold War and might cruelly be called a "branch office of Johns Hopkins."

Years later, in 2013, I became the chair of the board of what was by then called icddr,b. The centre has now published major vaccine trials led by Bangladeshi scientists in the New England Journal of Medicine , 1 and I was privileged to be on the steering committee of a major trial, also published in the New England Journal of Medicine , of a system for managing hypertension in rural Pakistan, Bangladesh, and Sri Lanka led by scientists from those countries. 2 Last year I was delighted to see a major trial of the polypill for the prevention of cardiovascular disease led from India and also published in the New England Journal of Medicine . 3

High quality research relevant to the needs of low and middle income countries is much commoner now than it was 40 years ago, and China has become a scientific leader. But, as the editors of this book describe in the introduction, there is still not nearly as much good research as there should be from the part of the world that carries most of the disease burden. Even worse, the medicine practised in some of these countries is disconnected from research and teaching, and driven more by profit than what is best for patients and the population. I have Indian friends who are terrified of seeing a cardiologist for fear that they will be given treatments they don't need.

I agree with the editors' diagnosis that "the main reasons for our having sunk into this deep morass is not because we are poor but because we have not intelligently examined, evaluated and investigated how we could use our own resources more effectively. We have tended to blindly follow what is being done in richer countries instead of trying to provide healthcare to our population which is accessible, affordable and, most importantly, appropriate even if this means deploying and working with informal healthcare providers."

Other people's research can be valuable, but it can never be as valuable as your own—addressing the problems that matter to your people with relevant methods and the tools you have. And we know that the very act of researching brings improvement, and (as I know to my cost) you can never learn about research from reading about it: you need to do it.

I've never quite understood why people in low and middle income countries would want to replicate the health systems of high income countries. Not only are those systems not relevant to the needs and circumstances of the low and middle income countries, but the systems in high income countries are increasingly unaffordable and unsustainable and not meeting the needs of their own populations.

Health systems in high income countries were developed decades ago and were designed to respond to the infectious disease and trauma that were then the main causes of suffering and death. Those problems could be cured, but now non-communicable disease is the main cause of suffering and death. Such disease cannot be cured and needs a different approach.

Non-communicable disease is now also the main cause of suffering and death in low and middle income countries (apart from some sub-Saharan countries, but even there it will soon be the main cause). The epidemiological transition happened very fast in low and middle income countries: in Bangladesh non-communicable disease caused about 10% of deaths in 1986 but nearer 80% by 2006. 4 I spent years working with 11 centres in low and middle income countries that were doing research, building capacity, and advising on policy in relation to non-communicable disease. We envisioned what a better system in low and middle income countries might look like—with an emphasis on public health, the social determinants of health, prevention, primary care, patient empowerment, and widespread use of evidence based guidelines. 5 (Such guidelines were developed by academics in South Africa as part of a package that allows good primary care where doctors are few or unavailable. 6)

We should have said more about the use of technology. Most people in low and middle income countries, even some of the poorest, now have mobile phones, which has meant that people can communicate without having to connect every house by wires, as happened with terrestrial phone systems in high income countries. Low and middle income countries can in this way "leapfrog" over a stage that was needed in high income countries, and the same can be done for health—not least by using mobile phones to provide access to care. Similarly, health systems in low and middle income countries might create health record systems where patients, not healthcare providers, own and control the records. Health systems in high income countries are just beginning to recognise the importance and inevitability of giving patients ownership and control of their records. (I have a conflict of interest here as I'm the chair of Patients Know Best, a company that gives patients in Britain and some other countries control of their records

Third cousin couples have the most children and grandchildren

Ed Yong · Not Exactly Rocket Science (Ed Yong archive)

Marriage between closely related cousins is a heavy taboo in many cultures and its critics often cite the higher risk of genetic diseases associated with inbreeding. That risk is certainly apparent for very close relatives, but a new study from Iceland shows that very distant relatives don't have it easy either. In the long run, they have just as few children and grandchildren as closely related ones.

Shuffling the genetic deck

Sex chromosomes aside, every person has two copies of each gene, one inherited from their father and one by their mother. Not every gene will be in correct working order, but there's a good chance that a faulty copy will be offset by a functional one from the other parent.

However, if two parents are closely related, there's a higher-than-average chance that they will already share some of the same genes and a similarly increased chance that their child will receive two defective copies. That can be very bad news indeed and in cases where important genes are affected, the results can include miscarriage, birth defects or early death.

Sex, then, is a shuffling of their genetic deck and theoretically the more closely related the partners are, the greater the chance that their child will be dealt a dud hand. And yet, some studies have found that some closely related couples actually do better than distant relatives in terms of the number of children they manage to raise. This trend is certainly unexpected and the big question is whether it is the result of biology or money.

Wealth or genes

In societies where close relatives marry, these unions tend to happen at a relatively early age and they provide avenues for families to retain wealth and land within bloodlines. These related couples enjoy the health benefits enjoyed by the rich as well as more time in which to raise a larger family. Together, these two effects could more than make up for any disadvantages wrought by their genes.

Earlier studies have done little to clear the confusion. They have mostly been conducted in parts of the world like India, Pakistan and the Middle East where marriage between close relatives is relatively frequent, but which are also home to enormous gulfs between the richest and poorest members of society. With demographics like these, sorting out the relative contribution of socioeconomics and biology is difficult.

To do that, what you need is a country with a small population where couples are reasonably closely related and with a very shallow gradient between rich and poor. Ideally, you'd also want this country to have excellent family records dating back several years. In short, you'd want to base your study in a country almost exactly like Iceland.

200 years of Iceland

Iceland is home to a tiny population of just over 300,000 people who enjoy a level of social equality that is almost unparalleled elsewhere in the world. Wealth, family size and cultural practices are fairly uniform. The country is also home to uniquely impressive geneaological records that allow today's Icelanders to track their family trees with exacting precision for centuries. These records are supplemented by thorough medical records and thousands of willingly donated genetic samples.

Agnar Helgason from deCODE Genetics , a pharmaceutical company located in Reykjavik, made good use of these records to study over 160,000 Icelandic couples since 1800. At this time, Iceland was still a poor agricultural nation and close-knit rural communities meant that on average, couples were related at the level of third or fourth cousins.

Since then, the country has prospered into a wealthy industrial one and the growing population has shifted to a mainly urban way of life. In doing so, people became more likely to find partners who were more distantly related and by 1965, couples were only related at the level of fifth cousins on average

As expected, Helgarson's study unveiled the dangers of close inbreeding. While the most closely related couples had the highest number of children, many of them failed to live long enough to have children of their own and in the long run these couples had the fewest grandchildren.

But surprisingly, distantly related couples were at a disadvantage too. In fact, Helgarson found that couples related at the level of third cousins eventually fostered the largest families. For example, among women born between 1800 and 1824, those partnered with men who were third cousins had an average of 4 children and 9 grandchildren, while those partnered with a distant eighth cousin had just 3 children and 7 grandchildren. For starting large families, very distant relatives were just as poor prospects as very close ones.

Over the 200 years included in the study, Iceland has seen a steep decline in both fertility and relatedness between couples. And despite all that, for every 25-year period the Helgarson looked at, the same pattern held – couples who were moderately closely related ended up with the largest number of descendants.

These remarkably consistent results have convinced Helgarson that the counterintuitive effect must have some biological foundation. Its exact nature will have to wait for another study and for now, we are left only with speculation.

It could be that a child's immune system may be more compatible with its mother's if its father is reasonably closely related to her. Alternatively, a union between distant relatives could serve to splinter groups of beneficial genes that have evolved in close association with each other.

Turning the tide: The Obesity Health Alliance's healthy weight strategy

julietwalker · The BMJ Blog

Tides are notoriously difficult to turn. But, when the time is right, even the strongest tides will turn—and now is one of those critical moments in relation to overweight and obesity in the UK. This follows 30 years of piecemeal action by governments to address a preventable health crisis that has become even more apparent during the covid-19 pandemic. Covid-19 has shone a light on the underlying health problems that make many of us more vulnerable when a new disease threatens population health and health systems that are already overstretched.

Recently the Obesity Health Alliance (OHA)—a coalition of 45 leading health charities and medical royal colleges —launched their new report "Turning the Tide: A 10-Year Healthy Weight Strategy'" for the UK. Obesity rates have increased significantly across the UK over the last 30 years, despite 14 government health strategies since 1991 that have set targets for obesity reduction in England , with similar failed efforts in the devolved nations. In total, these strategies contained 689 recommendations . Today, the majority of adults in England— 68% of men and 60% of women— are above a healthy weight, and over a quarter are living with obesity (27% of men and 29% of women). Excess weight prevalence is highest among the lowest socioeconomic groups in both adults and children.

The OHA's strategy was developed over two years by a working group chaired by Anne Johnson. Through a process of consensus-building, informed by a series of evidence reviews and expert papers, the strategy includes 30 recommendations for action in the years ahead.

It notes that the commercial food system has fundamentally changed over the second quarter of the 20 th century. The key goal of this food system was to eradicate hunger and ensure adequate food supply, however it has led to a complex adaptive system in which the need for readily available and accessible food has driven an explosion of ultra-processed foods that are highly palatable and heavily marketed to both meet and drive consumer demand . This has altered what we eat and how we behave in the same way that commercially mass produced cigarettes made smoking a normal part of life for the vast majority last century. Smoking resulted in hundreds of thousands of preventable premature deaths in the decades that followed until governments took action and changed the system to prevent smoking uptake and support existing smokers to quit.

The evidence is clear that the drivers of healthy weight are even more complex than those involved in tobacco use. The obesogenic environment , to which we are all exposed to from infancy onwards, whereby calorie-dense, nutrient poor food is readily accessible, abundant, affordable, and normalised, and physical activity opportunities are not easily available, affects us all. The pervasive and ubiquitous advertising and promotion of unhealthy food and drinks allows industry to "set the tone" of our food environment, creating, and maintaining this obesogenic environment that makes it inherently difficult to achieve a healthy diet.

Obesity cannot be addressed one person at a time, and behind the statistics are people living with obesity who often experience stigma and discrimination that can affect their mental health and willingness to access healthcare. Stigma stands in the way of public health and must be eradicated.

The strategy makes recommendations across key areas: 1) stigma; 2) food and drink products; 3) the marketing mix; 4) pricing; 5) the environment; 6) advertising and promotions; 7) early years; 8) management, treatment and support; and 9) improved policymaking processes. Each of these areas require addressing in their own right, but combined the recommendations set the clear, long-term, evidence-informed agenda needed to turn the tide and improve healthy weight across the UK population. The 30 recommendations made are set out in a "KIND" framework, aimed at building on existing policy progress as well as identifying new routes for action. The KIND framework includes:

Keep policies already in place or that are due to be implemented that supports a healthy weight environment;

Intensify existing policies or approaches to increase impact;

New proposals that are recommended for evidence-informed actions;

Develop policies based on the results of new, promising areas for research and investment.

The framework sets out not only the stage of policy development in which each recommendation sits, but also the actor(s) that should enact that recommendation. It is not enough to simply name route for action, but we need to establish clear directives for those bodies and organisations responsible to implement necessary change.

The role of industry is not absent here, and it is vitally important that we acknowledge the power that companies currently has over our food environment, as well as the regulation of the environment. We have clear evidence demonstrating that industry involvement in policy-making designed to remedy the harms industry products contribute to is not conducive to the creation of healthy public policy, and this must be actively managed throughout the policy process.

As we look ahead to recovery from the covid-19 pandemic, progressing efforts to achieve a healthy weight across the UK population must be at the heart of public health. It is time to move beyond individual-level policy recommendations, and implement wide-ranging changes to the system that got us here in the first place. This will not only save lives in the short and medium term, but will also serve to create a better and healthier future for our children and grandchildren, something they most surely deserve.

Dependency Injection in Typescript with tsyringe

GameChanger · GameChanger Tech

In any large object oriented codebase, managing dependencies can get difficult. Each class can require any number of third parties or other classes to function, and it can be hard to test the behavior of a single class with mocks if those dependencies aren't easy to provide.

Fortunately, there's a popular design pattern that can be applied to solve this problem, and that is dependency injection.

When using dependency injection, classes can be provided their dependencies through a constructor, and those dependencies can be swapped out easily for other implmentations. In tests, mocks can simply be substituted in to test class behavior.

While most of the time, this pattern is implemented with a framework, even without one manual dependency injection can give you some of these benefits.

Here at Gamechanger, we previously had a form of manual dependency injection in our typescript codebase. Each class would have a constructor that accepted its dependencies, which could be swapped out with mocks or a real implementation.

class BusinessLogic { constructor (dependencyA : DependencyA , dependencyB : DependencyB) {}

private void foo () { this . dependencyA . action (); } }

/\/ To instantiate this class, both dependencies must be created const businessLogic = new BusinessLogic (new DependencyA (), new DependencyB ());

/\/ to test this class, mocks can be passed in const testBusinessLogic = new BusinessLogic (new MockDependencyA (), new MockDependencyB ());

In this example, if you need to replace a dependency, you can just supply a different class in the constructor for BusinessLogic .

This can work great for a small number of classes with a tiny dependency tree, but as your codebase's number of dependencies grow, it can become difficult to manage. Once your dependencies have dependencies, its not as straightforward to get an instance of a class.

class DependencyC { constructor (dependencyE , DepedencyE) {

} }

class DependencyA { constructor (dependencyC : DependencyC , dependencyD : DependencyD) {

} }

/\/ Now, to instantiate BusinessLogic, we need to create a tree of instances. const dependencyC = new DependencyC (new DependencyE ()); const businessLogic = new BusinessLogic (new MockDependencyA (dependencyC , new DependencyD ()), new MockDependencyB ());

Even in this relatively mild example, it's starting to get complicated to manage the dependency tree. If you want to mock dependency C in the business logic dependency chain, you have to create all of the dependencies around it and pass those in.

When you need to test a particularly complicated class, setting up all its dependencies can take more time that writing the test itself! If you only need to mock a single subdependency, you need to instantiate everything all the way down until the mock is required, and then pass it in there.

Fortunately, there are dependency injection frameworks for typescript that can simplify the work that needs to be done.

Since we use typescript, we've moved to using https://github.com/microsoft/tsyringe

Tysringe allows you to tag a particular dependency as injectable with a decorator, and then very easily get an instance of it.

At its core, tsyringe provides you a dependency container that keeps track of all your dependencies. When you need to create an instance of a class, you can call resolve on the the container with an injection token and it will return you the right dependency registered under that token.

Our previous example becomes much easier to manage with this:

import { container , injectable } from ‘ tsyringe ‘ ;

@ injectable () class DependencyC { constructor (dependencyE , DepedencyE) {

} }

@ injectable () class DependencyA { constructor (dependencyC : DependencyC , dependencyD : DependencyD) {

} }

@ injectable () class BusinessLogic { constructor (dependencyA : DependencyA , dependencyB : DependencyB) {}

private void foo () { this . dependencyA . action (); } }

/\/ Now, all we need to do if we need an instance of business logic, is resolve it const businessLogicInstance = container . resolve (BusinessLogic);

That's it! All you need to do is tag your classes as injectable and tsyringe can take care of instantiating the whole dependency tree.

Tinder API Style Guide—Part 1

Tinder · Tinder Tech

Tinder API Style Guide — Part 1

Authored by: Nishant Mittal

Advisory Group: Vijaya Vangapandu , Devin Thomson , Serge Vartanov and Greg Giacovelli

API Style Guide Contributors: Nishant Mittal, Xing Wei and Felix Changoo

Over the past decade, Tinder has experienced exponential user growth. Today, the app processes over 1 billion Likes and Nopes per day . This rapid expansion has been accompanied by the introduction of numerous innovative experiences, such as AI Photo Selector, MatchMaker, Share My Date, Photo and ID Verification, Passport, and Explore, among others.

To support this scale and complexity, Tinder has undergone a significant organizational transformation. Initially, a few teams managed core services , but the company has since evolved into a structure with specialized, domain-centric teams . These teams include Trust and Safety, Identity, Engagement, Revenue, Recommendations, Chat, Profile, MatchList, Machine Learning, and Infrastructure .

As new teams were established to develop services and features within their domains, each team organically developed and adopted its own design and development practices. This diversity in methodologies led to material inconsistencies across services from different domains. Below are some of the issues discussed in detail.

Teams operating autonomously led to inconsistency in service communication. Some used standard HTTP status codes (2xx, 4xx, 5xx), while others returned custom error codes (example: 40001, 40301, 50001, etc) within a 200 status. This confused client applications, made error handling harder , and compromised the reliability of automated monitoring and logging systems dependent on standard codes.

Challenges in building collaboratively: We faced challenges in collaboration between backend and client teams (iOS, android, web). Backend teams often tailored schemas for their services when publishing new endpoints, making it hard for client teams to reuse them. Due to the delayed feedback loop , clients couldn't suggest schema changes, leading them to create their own model classes.

Lack of a Single Source of Truth : When clients maintained their own versions of data models, it led to the emergence of multiple, slightly different sets of models across the system. This fragmentation resulted in no single, authoritative source of truth, complicating maintenance efforts and slowing down feature releases. Additionally, the divergence between what was documented and what was actually implemented in each platform's model created inconsistencies, leading to potential bugs, miscommunications, and a less reliable system overall.

Other issues included adoption of inconsistent versioning strategies to release new versions of APIs, unpredictable ways of designing the URI depicting vague purposes of the API and poorly designed API contracts .

To address these inconsistencies, it became essential to lay down comprehensive documentation to standardize API development practices and enforce the adoption of these standard practices through manual and systematic audits.

Tinder API Standards

Almost all Tinder endpoints are designed as RESTful APIs (Representational State Transfer), allowing for scalable and flexible interactions across our platform. In this post, we'll explore the key standards we follow to create robust, RESTful APIs.

Designing URIs (Unique Resource Identifiers)

URI Pattern

URIs are essential in conveying the nature and behavior of an API, making it crucial to design them thoughtfully. We recommend following the pattern below whenever possible to ensure clear and effective URI design.

https://service-mesh-host-name/ [version]/[service-name]/[resource]/[sub-resource]

Note:

If the API has the same service name and resource, the URI should include only one (e.g., v2/themes , NOT v2/themes/themes).

The URI should clearly reflect the API's functionality . While the mentioned pattern is a good guideline, it's not a strict rule — it's a general approach that works for most URIs.

Path Prefix — Versioning

Versioning is a practice to manage changes to the APIs without disrupting the clients . It helps in clearly communicating the changes made allowing consumers to decide when to migrate to the latest version at their own pace.

Versioning Semantic and Strategy

At Tinder, we use versioning semantics with a path prefix (e.g., v1, v2). Our versioning strategy is to keep the last API version active until all clients migrate to the latest one. When a new version is published, create a sunset plan for the previous version , clearly communicate the timeline to clients, and ensure they are aligned with the migration schedule. Adjust the plan if needed, and continue supporting the last version until all clients have migrated.

When to publish a new Version of API?

Practising medicine in the virtual world—continuity of care is even more important

julietwalker · The BMJ Blog

If the covid-19 pandemic has taught us all one thing, it is the value of human relationships. More than ever before, doctors, nurses, and all allied healthcare professionals are working virtually via telephone and video consultations, changing the way that we relate to our patients. Guidelines have sprung up telling us how to do this better, but something fundamental is missing: continuity of care. [1]

The importance of continuity of care has been amplified for me during the pandemic, both as a patient with type 1 diabetes for 35 years, and as a paediatric diabetologist. As a patient, I don't want to have to repeat my story every time I see a new doctor, especially the difficult things that need discussing. As a doctor, how do I learn whether the advice I gave was correct, and how do I find out what my patient really wants from me, or when to gently nudge them in a different direction or provide reassurance? Knowledge of the patient allows me to be a better doctor, to give an individual approach, rather than responding to a blood glucose value.

Plenty of evidence exists about the benefits of continuity of care across many health conditions and ages. It improves satisfaction, reduces unnecessary referrals, and lowers mortality. [2,3] Continuity will matter more to some patients (and doctors) and more in certain conditions, than others. Some will sacrifice continuity if it means a long wait. Others who find the relationship therapeutic as much as the medicine, will rank continuity more highly. Managing and understanding that must be central to the discussion.

A snapshot online survey of 80 paediatric diabetes consultants in the UK, collected for a debate on continuity of care, during the British Society of Paediatric Endocrinology annual conference , identified that only 47% of consultants are able to practise continuity of care, despite the acknowledgement that it was important for their continued professional development and job satisfaction, and for their patients.

So if continuity is known to be beneficial, why is it not practised more often? One barrier is infrastructure and training: how do junior doctors get experience without moving to different specialties and clinics? If consultants need to be elsewhere, like a ward round or on holiday, should the clinic be cancelled? All this needs to be worked out and there isn't a one-size-fits-all solution.

Many healthcare professionals have had to set up virtual clinics overnight. This drive towards innovation is a good thing: it has cut bureaucratic processes, can save money for the health system, and save time for the patient. As we hopefully start to emerge from the pandemic, the NHS and other healthcare systems will need to redefine how we function. Virtual working will be on the table, but before setting it up indiscriminately, we need to pause, and consider how it can best be used. The patient should remain at the centre, and be offered choice, to maintain and improve patient satisfaction and outcomes. Patient continuity, especially for chronic conditions, should not be unwittingly sacrificed: forming a healthy attachment between doctor and patient is at the core of how we relate and trust.

If the pandemic has taught us about the importance of human relationships, it has also taught us that if there is enough will, a solution can be found.

Rachel Besser is a consultant paediatric diabetologist at Oxford University Hospitals NHS Foundation Trust and author of Diabetes Through the Looking Glass – seeing diabetes from your child's perspective . @BesserBesser

Competing interests : none declared .

References:

Car J, Choon G, Koh H, Foong PS, Wang CJ. Video consultations in primary and specialist care during the covid-19 pandemic and beyond BMJ 2020; 20;371:m3945.

Pereira Gray DJ, Sidaway-Lee K, White E, Thorne A, Evans PH. Continuity of care with doctors—a matter of life and death? A systematic review of continuity of care and mortality. BMJ Open 2018;8:e021161.

Helen Salisbury H: Measuring continuity of care BMJ 2019 ;3;367:l6567. doi: 10.1136/bmj.l6567.

An electric day 3 of TEDNext 2025

Brian Greene · TED Blog

TED’s Monique Ruff-Bell (left) and Helen Walters (right) host Session 5 of TEDNext on November 11, 2025, in Atlanta, GA. (Photo: Ryan Lash / TED)

For day 3 of TEDNext 2025, two magical sessions of talks put forward ideas on how to grow thoughtfully, think radically and redesign a better world. With stories from our AI-powered present (and future) as well as a healthy dose of ingenious human problem-solving, the talks of day 3 surprised, delighted and shed new light on what it means to be human right now.

What exactly is TEDNext? A vibrant, three-day exploration of what’s next, propelling the “future you” to think expansively at every level, from personal to global. The second-ever TEDNext conference, held in Atlanta, continues an expansion of the annual slate of conferences from TED, with a conference designed to spark imagination, embrace possibility and foster dreams about what the next version of “you” can be.

Watch TEDNext 2025 on TED Live , check out more photos from the event and learn more about attending a future TED conference .

Some key takeaways from day 3:

Amanda Montell speaks at TEDNext on November 11, 2025, in Atlanta, GA. (Photo: Ryan Lash / TED)

The line between awe and indoctrination — and how to actually seek happiness. Writer and podcaster Amanda Montell thinks we’re living through the “cult-iest” era on record, as the hard-wired cognitive biases that helped early humans survive brush up against Information Age tools and communication tactics. She shares how to recognize the strategies and language tricks cults use to coerce — from thought-terminating clichés designed to shut down independent reflection to loaded buzzwords that feel like enlightenment — as well as some crucial tips to help you preserve the ability to think for yourself. Approaching our need for fulfillment from a different angle, happiness scientist Sonja Lyubomirsky distills the essential lessons from scientific research into humanity’s most sought-after emotion: happiness. Her number one hack? To approach conversations with others as experiences designed to take walls down, not changes to share your highlight reel. By sharing deeply and listening to learn, Lyubomirsky thinks we can all unlock the potential for happiness inside of us.

Nayeema Raza speaks at TEDNext on November 11, 2025, in Atlanta, GA. (Photo: Erin Lubin / TED)

Some pro-human takes on AI and algorithms. How do we stay “real” when algorithms and chatbots are constantly influencing our behavior to the benefit of platforms like ChatGPT, Spotify, X and others? Etymologist and content creator Adam Aleksic sounds the alarm on how AI tools are changing how we communicate — down to our very word choices — and, possibly, rewiring the underlying patterns of our thoughts, encouraging us to remember that these emerging tools aren’t neutral. Self-described “dumb questions” advocate Nayeema Raza picks up the thread of AI tools hijacking our lives — not with an anti-tech tech, but with a pro-human one. She makes a case for the return of three old habits — to pause and notice our urge to reach for our devices, to live in wonder and to ask questions out loud instead of in private — to reconnect with what actually matters in our lives.

John Mills speaks at TEDNext on November 11, 2025, in Atlanta, GA. (Photo: Erin Lubin / TED)

Tech meets the natural world. After watching a series of wildfires rage around his home in Northern California in 2019, civic tech pioneer John Mills had a simple question: Where is the information? With firefighters relying on communications technology from the 1930s, often the answer was: too little, too late and too sporadic. Enter Watch Duty, a non-profit alert system Mills created with the help of a few radio operators and volunteer engineers from Silicon Valley. Developed in an 80-day sprint, Watch Duty beat government alert systems by nearly an hour just a handful of days after it launched, and it has continued to provide residents and first responders with the life-saving heads-up they need to escape danger, proving that a small group of thoughtful, committed citizens can indeed change the world. Telling another story from the positive side of tech, ocean explorer Eric Stackpole takes us into the murky depths to explore the mysterious worlds of sperm whales. Sequestered aboard a ship filming a show for National Geographic during the early days of the COVID pandemic, Stackpole and his colleagues concocted a “very maker-y” camera rig that ended up revealing a previously unreachable world — capturing footage of sperm whales communicating and coordinating for the first time on camera. “The question isn’t: ‘What can we explore?‘” Stackpole says. “The real question is: ‘What will we wonder about next?’”

Manoush Zomorodi speaks at TEDNext on November 11, 2025, in Atlanta, GA. (Photo: Erin Lubin / TED)

New thinking on pain, the mind and body. We’ve let our screens quietly train our days into locked-down stillness, says journalist and host of NPR’s TED Radio Hour , Manoush Zomorodi , and it’s led to many of us feeling exhausted most of the time. She makes the case that tiny, regular bursts of activity baked into our days — known as “movement breaks” — can flip the body’s switches, brightening mood, sharpening focus and even regulating blood sugar. That same spirit of rethinking the obvious extends to how we think itself: in a personal talk exploring the mind’s eye, puzzle wizard and editorial director of TED-Ed Animations Alex Rosenthal shows how our inner worlds can vary wildly, and that when we build teams across those differences, creativity and problem-solving multiply. Taking that openness into the clinic, Dr. Sanjay Gupta explores new thoughts on pain — how it isn’t just a signal in tissue but a story shaped by environment, history, sleep and support. He reveals how

Apache Airflow on AWS ECS

GameChanger · GameChanger Tech

An overview of what AWS ECS is, how to run Apache Airflow and tasks on it for eased infrastructure maintenance, and what we've encountered so that you have an easier time getting up and running.

Our data team recently made the transition in workflow systems from Jenkins to Apache Airflow . Airflow was a completely new system to us that we had no previous experience with but is the current industry standard for the sort of data-centric workflow jobs we were looking to run. While it hasn't been a perfect fit, we have been able to get a lot of benefits from it: jobs are defined in code, we've the history of each job, it goes through our normal pull request process, and everyone on the team is able to read and write jobs.

Since our team is data focused, we wanted our Airflow setup to be as easy to maintain as possible, especially around infrastructure, so we have minimal distractions with high resiliency. This led us to using AWS ECS not only to run Airflow but for our bigger tasks that are already containerized. Not familiar with ECS? Or how to run Airflow or its tasks on it? Don't worry, we weren't either.

This is what we've learned.

tl;dr Running Airflow's infrastructure on ECS is super easy but running the ECS operator needs hecka help setting up.

What is ECS?

ECS is AWS's Elastic Container Service , designed to let you run containers without worrying about servers. You do this by creating a cluster for your system, define task definitions for the tasks you want to run, and possibly group your tasks into services. You can also choose if you want to have your containers fully managed (ECS on Fargate), kind of managed but kind of not (ECS on EC2), or if you'd like to use Kubernetes.

If all of that made sense to you, congratulations: you can skip to the next section! If you need a bit more of a breakdown, here's what you need to know:

A container is like a snapshot of a tiny computer that you can run anywhere. If, for example, you have some Docker containers to run, you can run them on my mac Mac, my father's Mac, my mother's Windows, my sister's Linux, a server in a public cloud, a server in a private data center with an undisclosed location for security reasons — anywhere there's a computer that supports the container's setup can run that container, and you'll get the same result from the container every time.

The main advantage containers give us is that it simplifies making sure that something running locally on an engineer's machine, and running perfectly fine in a staging environment QA is testing in, will also run in a production environment where users are making requests, since they're all the same container being run the same way. No dependency went missing in the middle, no permission was accidentally changed locally that hid a problem: it's the same thing running the same way.

You might be saying, that sounds great! I'mma launch containers everywhere! And that is the attitude ECS is meant to capture and bring to fruition, because your other option to running containers is a bit more hands on: after all, a container needs to execute on a system, and that means hardware. That means a server somewhere you define and setup with an operating system of its own, and you might have to install the container's setup system on it (for example, installing Docker), and you have keep it all up to date, plus you want to secure it, and you have to have a way to get containers started on that server, and you probably want a way to monitor for if a container goes down with a way to restart it, and there's security patches.... This is starting to be a long list, and while it's been a pretty standard list for a while, we deserve better systems .

We're a data team, after all, and none of that is really about working with our data.

That's where ECS comes in. Instead of running servers (EC2) where you do all the above things to to get your containers running, you can use ECS with Fargate to worry only about the containers: here's the CPU and memory it needs, here's the start command, here's a health check, give it these environment values, and I want two of these running at all times. Boom: your containers are running, and restart as needed, and things are kept up to date for you, and there's monitoring built in.

There's a lot more nuance that goes into picking if you want to run ECS on Fargate or EC2 or Kubernetes, but if you're still reading this, you probably want Fargate: with Fargate, you only need to worry about your containers, and the rest is taken care of for you.

Now that we're all caught up...

ECS with Terraform

As discussed in a previous post , we're huge fans of using Terraform for our team's infrastructure, especially modules to keep systems in line across environments; I'll let you read that post if you'd like some background knowledge on how we use Terraform modules and have them interact with Consul for sharing values. The short story though is, as with the above, it keeps the setup easy for our data team so we

The AI boom has plunged a small Pennsylvania town into chaos

Rebecca Egan McCarthy · Grist

“I don’t like to see anyone upset,” said Nick Farris of Provident Real Estate Advisors. He was sitting in the front of a crowd of roughly 150 inside Valley View High School’s auditorium in Archbald, a town of about 7,500, huddled between two mountain ranges in Pennsylvania’s Lackawanna Valley. Farris was there to represent the developer for Project Scott, one of many data center campuses coming to town. “However,” he said. “I think that this is the best data center site in this area of the country, by far.” The audience had been fairly quiet, bundled in thick coats against the late January cold. But as Farris spoke about data centers as a boon for communities, they began to laugh, drawing a rebuke from town officials.

“What about the children?” someone shouted from the crowd. The children were watching from the walls; long banners of Valley View Performing Arts students hanging around the auditorium like championship pennants. Project Scott and four other data facilities will sit just a few thousand feet from the middle and high schools.

“Isn’t there a missile plant next door?” Farris said, getting aggravated. He was referring to Lockheed Martin’s 350,000-square-foot Missiles and Fire Control facility directly next to the high school, parts of which are highly contaminated .

“That sucks too!” another attendee yelled back. This was nothing to worry about, Farris tried to convince the audience. This would bring in tax revenue, he said. It was just an office park, albeit one with roughly 450 diesel backup generators.

“It’s going to be away from everyone,” Farris kept repeating, to rising jeers. He was wearing a knit turtleneck with a large American flag emblazoned across the chest. “It’s not going to bother anyone.”

The specifications say something different: Five developers are planning to build six data center campuses in Archbald, which will cover a full 14 percent of the town, evict a trailer park, and border many residential properties. One campus alone, as The Scranton Times-Tribune reporter Frank Lefneskey pointed out, is expected to use more power than the region’s largest power plant is able to produce.

Pennsylvania has become an epicenter of the data center boom in the United States, with over 50 campuses in development. Eleven of them are slated for Lackawanna County alone. Archbald, with six campuses composed of 51 massive buildings, has the most of any municipality in Pennsylvania.

Despite the public outcry, it has been surprisingly difficult to learn what Archbald’s elected officials think of the massive industry moving in. None of the town’s seven council members responded to my emails, so I stopped by the borough administration building in person a few weeks before Christmas, and was told to wait in the lobby while they held an informal closed meeting in council chambers. When the door opened, it was clear that anyone who actually had the power to make or break the data center plans had quietly filed out the back, leaving me with Archbald’s unelected borough manager, Dan Markey, who essentially runs the town, but cannot vote for or against development.

What did he think about artificial general intelligence, or AGI — the idea that eventually these efforts will produce something like a “computer god” capable of solving climate change, ending hunger, revealing the full breadth of science, and performing any number of other miracles? “I believe in one God, and it’s not a computer,” Markey told me evenly. By now, hundreds of towns across the country have been caught up in the rush to build large language models that can parse unfathomable amounts of data to write copy, answer any query, develop new vaccines, and likely render a large number of jobs obsolete. That rush accelerated into an all-out arms race over the past year. What it means in practice is an enormous amount of data centers — enough to approximate a minor deity and enough, as OpenAI cofounder and former chief scientist Ilya Sutskever once speculated, to “cover the Earth.” Markey told me he had ChatGPT on his phone, but didn’t use it much.

“I don’t think anyone in their right mind wants to see the world covered in data centers,” Markey said. “[But], according to Pennsylvania law, we have to have a zone for everything. An adult bookstore, a strip club, a concrete and asphalt plant — anything that wants to come here. We have to have a zone for it. If it’s not zoned, it’s allowed to go anywhere.”

In most states, towns have the right to exclude businesses that they find disagreeable — a wealthy suburb, for example, would likely reject a landfill or gas plant — but in Pennsylvania, towns must allocate some patch of land to these “undesirable industries.” Some municipalities deal with this by forming what are called zoning collaboratives, which allow them to plan as a region for pollution. One town might get data centers, another gas plants, another a landfill. Markey approached the nearby boroughs of Blakely and Dickson City to discuss the possibility, but the data center development rush has outpaced him.

Whatever’s attracting data centers to the area, it’s forced Markey to answer for a rush of development, unprecedented since the town was settled in the 1840s in service of an industry that will bring vanishingly few jobs. Residents kept bringing up the threat of collapse — the network of empty mine shafts running underground, the town beneath the town, the structural instability that would accompany these massive buildings. The data centers would drive bears into town, they said, rattlesnakes into yards, and without trees to stabilize them, the mountains themselves would begin to crumble, sending landslides into the valley.

“I just try to listen, and I try to separate the valid concerns from the things that just sound like the sky is falling,” said Markey. “I was approached at a gas station a couple months ago and was told that I was going to kill everyone who lived

Building Obsidian, Tinder's Design System

Tinder · Tinder Tech

Author: Darragh Burke, Software Engineer II, Web Development at Tinder

Tinder's UI Opportunities: Wildfire

When Tinder first launched in 2012, it pioneered a brand new user experience: the Swipe Right® and "Swipe Left"™ features. The app's simplicity was a big part of what made it so appealing.

An early version of Tinder

We've emphasized building new features and moving fast while growing, but a decade later, Tinder has become a significantly more established app. It's become harder to maintain consistent design practices across every screen and component and the codebase suffered too.

The cost of designing, building, and maintaining features was becoming unsustainable. We needed a way to unify the appearance of the application, save time for product managers, designers, and engineers, and deliver a more consistent appearance to our end users. We needed a design system .

What is a design system?

Using patterns in designing software isn't a new idea — it's been around since at least the 1960s. But it was Brad Frost who popularized the phrase "Design System" with his 2016 book Atomic Design .

A design system is essentially a standardized collection of styles and reusable components that dramatically reduces the cost of design and development and increases the quality of an application.

One common idea across many design systems is the notion of "Design Tokens". These are tiny bits of design data that can be used to build interfaces. Design Tokens can consist of colors, gradients, font faces, border radii, or any other value that's used in a UI.

We think of design tokens in two categories:

Base Tokens . These directly reference styling data like colors or font sizes, and might be named something like "Blue 15" or "Brand Primary". The name describes the value inside. We don't think these should be used in designs directly, as they contain little semantic information about the purpose of the token.

Context Tokens . These tokens reference other tokens instead of raw values directly. They contain information about where they should be used. They can also contain theming information — they can map to a different token depending on the theme the user is in, such as light mode or dark mode. For example, a token called "Text Primary" might map to Black in light mode and White in dark mode. This would result in dark text in light mode and light text in dark mode.

By using context tokens to build interfaces, it becomes trivial to enable dark mode or even other types of themes. It also means that all styles are controlled from a single place, so it's easy to update a style and see it reflected across the entire application.

Using standard tokens also lets us craft styles carefully for accessibility. For example, we can define sets of matching text and background colors that will make our application more accessible to those with visual impairments. By using our defined pairs of background and foreground tokens, consumers of the design system can feel confident that their designs will meet contrast requirements.

Finally, tokens help to align design and engineering on a common language for talking about styling. Rather than describing "that black text color", we can simply reference the "Light/Semantic/Text/Primary" token. This resolves ambiguity and ensures everyone is on the same page. If we think of a user interface as a cooking recipe, then design tokens are the ingredients — and it's much more helpful to call for "butter", a concept everyone understands, rather than "triglycerides".

Another core feature of a design system is the idea of reusable components. These are UI elements such as text inputs or checkboxes that have a predefined set of styles and can be dropped in anywhere. Designing and building reusable components can be a huge time-saver in the long run, reducing the amount of work for both designers and engineers to build features.

There were several attempts to introduce these ideas to Tinder over the years, but they always came up a little bit short. There was no single, cohesive design system for Tinder until the UI Platform team was officially formed in 2020. This was a team whose explicit purpose was to create a set of design standards that could be used to build features at a faster pace, with more consistency and accessibility. Ultimately, the vision we settled on was this:

> *Style Dictionary then generates artifacts for each platform.*
>
> Tinder

"Empower our designers, product managers, and engineers to move fast at scale while reducing operational cost of creating new experiences."

We built Obsidian to realize that vision.

Obsidian is Tinder's Design System. It's an entire design ecosystem composed of a set of standardized design tokens, reusable UI components, documentation, and design tools. It exists to resolve ambiguity in the design process.

Healthcare in our hands: Putting babies, children, and young people at the centre of their care

kellybrendel · The BMJ Blog

Everyone is entitled to a say in the healthcare services they use and a child or young person is no different, say Aishah Farooq, Emma Beeden, and Catherine White

Accessing healthcare services can be daunting at any age, but even more so as a baby, child, or young person. Can you imagine being told life changing information and not having any idea of what's going on? It's a difficult and helpless position to be in, and we felt overwhelmed, anxious, and vulnerable when we experienced this as children and young people. When all your decisions are made for you, albeit with the best of intentions, and you don't have much involvement apart from receiving the treatment, it's easy to feel fearful and powerless. Yet everyone is entitled to a say in the healthcare services that they use and a child or young person is no different.

The United Nations Convention on the Rights of the Child states every "child who is capable of forming his or her own views has the right to express those views freely in all matters affecting the child." The National Institute for Health and Care Excellence (NICE) recently published guidance on Babies, Children, and Young People's Experience of Healthcare, and this principle has been embedded throughout the guideline. By actively involving the views of children, young people, and the parents and carers of babies and young children, this guidance truly represents those who use the service and will mean it can make a real difference.

The recognition that children and young people have a right to share their perspectives is not only woven throughout the recommendations, but was also remembered all the way through the guideline development. From scope development to publication, young people with personal experiences of accessing healthcare have been involved in this guideline.

Having personal experience of accessing healthcare gives you a unique viewpoint that is not always heard, especially if you are from under-represented groups, including those who are under 18, disabled, from an ethnic minority background, or who don't have English as a first language. It was vital for these voices and experiences to be heard, listened to, and then put into this guidance to ensure it can improve care for as many people as possible. The guideline committee included four young people with varying levels of experience accessing healthcare (including two of us, Aishah and Emma). Having this wealth of experience to draw on meant that we were able to bring our expert opinion and real life testimonies to the guideline committee meetings, rather than just relying on examples from research or journal articles.

The guideline has formed recommendations in a broad range of areas, but we want to highlight two in particular. First, communication. We know from our experiences that often when our parents and carers are present, healthcare professionals will direct all communication towards them, even though it's about us. This can be disempowering and risks us disengaging with our healthcare (for example, by being reluctant to access services or adhere to treatment). It's essential that we feel included in conversations and that our preference of communication is understood.

It is so important to engage with and provide information to a child or young person in a form of communication that they feel comfortable with, whether that be through Makaton, pitching the language used at the right level, or using additional visual aids for support. It can help them feel more at ease, builds trust, and opens up more opportunities for them to be involved in their own healthcare journey.

> *At this age we can have little involvement and voice in our healthcare, but shared decision making helps to position us in the centre of our care.*
>
> kellybrendel

Shared decision making is an important part of any individual's healthcare journey—especially a child or young person's. At this age we can have little involvement and voice in our healthcare, but shared decision making helps to position us in the centre of our care. The NICE guideline sheds light on how children and young people can be supported to make decisions, even if they're as simple as choosing a plaster cast colour or food options on a menu. As a child and young person, we've both had similar options chosen for us, but when we knew what we wanted, why weren't we asked? Being involved in simpler decisions that may not have a direct impact on our healthcare outcome can seem unimportant, however it's crucial, because it affects our overall perception of healthcare and builds our confidence to engage in bigger healthcare decisions. It is also well known that the transition from paediatric to adult services can be a really difficult time for patients and it's not hard to see why: how can we be responsible for our own care if we haven't had the opportunity to learn how to do it?

As part of the evidence for our recommendations, NICE commissioned the National Children's Bureau to run focus groups with 200 children aged 4-15 years old. To conclude, we'd like to include their voices about how they felt being involved: "Young person's voices should be listened to, valued, and considered to ensure that the health service can be accessible to all"; "The best thing is giving our opinions and being part of something important"; "I like you listening to my views, it's awesome!"; "I really like that it feels that you all care about my opinion."

Through this guideline, we saw that it was possible to fully engage with children and young people. Now it's time to put that into practice across healthcare.

Aishah Farooq, committee member of the NICE guide-

Jeffrey Aronson: When I Use a Word . . . Snowflakes

jross · The BMJ Blog

I love snowflakes. I enjoy crunching them underfoot on a crisp winter's day and the silky feeling that you get when skiing through a fresh fall.

The word "snowflake" entered English in the early 18 th century, first recorded in a pantomime called Cupid & Psyche, or, Colombine-Courtezan, based on the tale told by the Numidian writer Apuleius (c124–c170) in The Golden Ass: " Soft as the cygnet's down his wings, And as the falling snowflake fair". This was a surprisingly late entry, considering how early its separate components, "snow" and "flake", had appeared. The earliest instance of "snow" cited in the Oxford English Dictionary is in the Vespasian psalter , from c825 AD: " Se seleð snaw swe swe wulle ", which is an Old English translation of the Latin version of Psalm 147, verse 16: "qui dat nivem quasi lanam", or in the original Hebrew

which literally means "the giving of snow like wool", more clearly translated as "he spreads snow like a woollen blanket". "Flake" first appears, linked to snow, in Chaucer's House of Fame (c1384): "As flakes fallen in great snowes".

But words evolve and, like many other words, "snowflake" has accrued many other meanings.

Snow-buntings have white bodies and what appear to be flecks of snow on their wings. Starting in the late 17 th century, they were therefore called snow-flecks. Then, in the late 18 th century, that became corrupted into "snowflakes".

The snow-bunting, snow-fleck, or snowflake (Wikimedia commons).

At about the same time, William Curtis published his Flora Londinensis , which included the picture of a purple foxglove that William Withering used as the frontispiece to his 1785 monograph on digitalis, An Account of the Foxglove . Curtis gave the name snowflake to the Leucoium aestivum , to distinguish it from Galanthus species of snowdrops, both members of the family Amaryllidaceae . These plants contain galantamine , a cholinesterase inhibitor that has been used to treat dementias; the results have been unimpressive in both Alzheimer's disease and vascular dementias .

As in the psalm, snow and wool also come together in the use of the word "snowflake", when it describes a method of weaving woollen cloth.

A snowflake is also the name given to a hairline crack in metal, arising when the metal has not been allowed to cool down sufficiently slowly, and described in a 1919 issue of the Bulletin of the American Institute of Mining Engineers as a "white silvery area, which always has the appearance of being of a very coarsely crystalline structure".

Snowflakes appear in mathematics too, and I discuss one instance in the Interesting integer section below. Here is another. Draw an equilateral triangle (Figure a; black). Now on the middle third of each side draw another equilateral triangle and erase its base (b; red). Now repeat that on each available side in the new shape (c; blue). If you continue doing this you get a shape that resembles a snow flake (d). This is a fractal pattern, one that has the same local characteristics no matter how high a magnification you choose to look at it. The boundary is of infinite length, because you can go on adding new triangles forever. However the area it encloses is finite and is 60% larger than the area of the original triangle.

One more meaning of "snowflake" remains to be explored.

Ah, well, I should have expected that to happen

Finally, an aphorism from a book called Myśli nieuczesane nowe (More Unkempt Thoughts , 1964) by the Polish poet Stanisław Jerzy Lec: Żaden płatek śniegu nie czuje się odpowiedzialny za lawinę. No snowflake feels responsible for an avalanche.

Jeffrey Aronson is a clinical pharmacologist, working in the Centre for Evidence Based Medicine in Oxford's Nuffield Department of Primary Care Health Sciences. He is also president emeritus of the British Pharmacological Society.

Competing interests: none declared.

The post Jeffrey Aronson: When I Use a Word . . . Snowflakes appeared first on The BMJ .

Climate emergency: it is time for the aggressive, life saving interventions

julietwalker · The BMJ Blog

In the face of a devastating potentially fatal diagnosis, healthcare professionals and families often turn to palliative medicine for guidance and support. It is not about "giving up"—and in many situations, especially in paediatric medicine—palliative care can and should be provided alongside active treatments. A child with advanced malignant disease, for example, may have chemotherapy aimed at cure, but will also benefit from the holistic and symptom management approach that palliative medicine can provide alongside. Both the aggressive treatment and the palliative management will contribute to healing.

Our planet is in need of palliative care. For years, climate scientists, effectively part of the planet's "medical" team, have urged us to change the way we live in order to slow, or even reverse, our planet's rapid spiral towards death. They have shown us the evidence, given us numbers, predicted time scales, and calculated the positive effects of treatment. We have ignored their advice and have continued to do the things that exacerbate the symptoms, that make the planet more unwell, and hasten it's death. Some have hoped that somehow the "medics" have got it wrong. Others perhaps have taken the approach that if we don't think or talk about it, then it won't happen—a common approach to death throughout our society. Whatever our reasons, we have ignored the symptoms and professional advice for so long that the planet is dying. And this will not be a painless death, surrounded by loved ones and with medication to ease the suffering. This will be a slow and painful death, of flooding, of drought, of starvation, of rising temperatures, of forest fires, of increasing pollution, of increased disease, all bringing emotional suffering and pain on a huge global scale.

It is time for the aggressive, life saving interventions. There is still the hope of cure, and we know what treatments will work. We need big changes, such as a rapid transition away from fossil fuels and investment in renewable energy. We all need to embrace the "treatment" in the knowledge that it can be life-saving. But alongside this we must not ignore the importance and impact of a palliative care approach. In the face of death, every day matters and every person matters. Not just the dying person, but those close to them. We teach our medical students that palliative care is everyone's responsibility—every single person that comes into contact with the patient has a role to play. It may be just noticing that a water jug needs filling or sitting to listen, but you only have to hear a patient's story to know the huge impact these apparently small acts of kindness, thoughtfulness, and compassion can have.

Our approach to the planet must be the same. In the face of death, every day matters. Every single person who lives on this planet, who breathes it's increasingly toxic air, matters. Small acts of kindness, thoughtfulness, and compassion towards our beautiful planet will have an impact. We cannot all roar like Greta Thunberg, but we can all be the person who shows up every day, with the seemingly small acts that matter. We must follow the "medics" advice and stop doing the things that will exacerbate symptoms and contribute further to an early demise. We can all buy less, buy wisely, buy second hand. We must all learn to throw less away, repair, reuse things that others no longer need, and borrow things we don't need often. We can reduce our use of "disposable" items by switching, for example, to cloth nappies and re-usable menstrual products. We need to change our diet to one that promotes healing, reducing our meat consumption (or even going veggie), and thinking about alternatives to cow's milk. We need to make life style changes if we want to live. And contrary to what some may think, these will be life style changes that improve our quality of life as well as promoting our survival.

We can no longer ignore the reality that is facing us. As the planet moves closer to inevitable death, we must hope that the aggressive treatments needed will come, and that these will not come too late. And we must embrace and commit to them, with the hope they bring. But none of us should sit idly by, waiting for the cure. We must all adopt the palliative approach, which will help to nurse our planet back to health. Every act matters. Every person counts. We must all stop feeding the disease and be part of the cure.

Finella Craig is a consultant in paediatric palliative medicine in London. She travels everywhere on her bike, including to all her patient home visits, and is taking part in Ride for Their Lives London to Glasgow.

Competing interests : none declared.

We must protect our planet for our children's future

julietwalker · The BMJ Blog

The defining characteristic of child abuse, be it physical, emotional, or through neglect, is that the child suffers significant harm. At its extreme child abuse is fatal. As a species, through neglecting the ecological determinants of health we have killed millions of children and are in the process of killing many thousands more, while condemning millions to ongoing physical and emotional suffering.

Already, 920 million children suffer from water scarcity, with numbers rising . By 2030 there may be more than 100,000 additional deaths in children under five due to malnutrition attributable to climate change, and an additional 7.5 million children with moderate to severe stunting of their growth . Children have suffered and died due to extreme weather events—floods, cyclones, heatwaves, and consequent wildfires. Climate change is now a leading cause of forced migration, disrupting children's home and family lives, education, and healthcare. Hundreds of thousands of children under the age of five die as a result of ambient air pollution each year, and hundreds of thousands more due to household air pollution .

Breathing faster, children inhale more of any pollutant per unit of body weight than adults. Their developing organs are particularly vulnerable in the womb and in early life. Recent estimates suggest exposure to air pollution during pregnancy may be responsible for nearly three million low birth weight and nearly six million premature births per year. Low birthweight and prematurity are key risk factors with potential irreversible impacts on health and wellbeing throughout life. This is significant physical harm on a global scale, with no safeguarding panels, or child death reviews.

There are emotional impacts too. While having limited agency to change the trajectory of climate change themselves, children and young people will live to see more of its impact than those of us already a fair way through our lives. Anxiety is a rational response.

We know from surveys that many thousands of young people are feeling sad, anxious, angry, powerless, helpless, and guilty about climate change . "How dare you?" roared Greta Thunberg in 2019, "you have stolen my dreams and my childhood."

Ella Kissi-Debrah isn't roaring. She died in 2013 from severe asthma, exacerbated by her chronic exposure to high levels of air pollution in her neighbourhood by London's South Circular road. We must roar on her behalf, and on behalf of all the other millions of children whose lives we are destroying. Call the global social services department and report ourselves for failing to change our ways, even when we knew we should. We are all perpetrators. At home, at work, weekdays, weekends, holidays, celebrations, with families, with friends—consuming, consuming, consuming. Depleting the global commons of natural resources our children will need, and increasing the burden of pollutants in our air, our water, and our land. We do this directly—sometimes burning fuel for the most frivolous of reasons, such as to avoid wearing another layer of clothing, or for our car to make the right vroom vroom noise. We do it indirectly too —we have learned to derive gratification from acquiring products whose manufacture, packaging, distribution, and disposal depletes and pollutes beyond our field of vision. Shortage of toys for Christmas? Fantastic—let's do what genuinely makes us happy instead.

We have to change. We have to tackle climate change bottom up, top down, any which way we can. Child protection is everyone's business, and climate protection is child protection. Climate risk assessments and mitigation protocols must be ubiquitous—every business, service, school, university, sports club, choir, band, event and so on should have these and be reviewed and audited against them.

Health professionals, who fully understand the implications of destroying the ecological determinants of health, must lead from the front. People noticed when we all stopped smoking. They will notice when we take climate change seriously.

The UN must be our global safeguarding committee. Each Conference of the Parties should include a child protection meeting, at which data and narratives on child suffering and deaths due to climate change are reviewed. At COP26 we must all be held to account.

There is no place of safety to which we can remove our children — no alternative world where climate change has been arrested and there is safe air, water, soil, food, and shelter for all. Everyone must play their part in rescuing this world to make it safe for children, and I for one will continue to roar until we do.

Lucy Reynolds is a community paediatrician living and working in Glasgow. Through the climate change and child health webinars run by the International Society for Social Paediatrics and Child Health she became involved in the Ride for Their Lives initiative and the RCPCH climate change working group.

Competing interests : none declared.

The post We must protect our planet for our children's future appeared first on The BMJ .

Learning Kotlin Constructor as a Java Developer

GameChanger · GameChanger Tech

I have been developing Android apps in Java for years. I recently joined GameChanger and was excited to learn that GameChanger is using Kotlin. I originally thought that moving to Kotlin would be as simple as learning some new syntax, but I discovered that there was more to it. After a brief learning period, I'm up and running working in Kotlin. In this article, I'll save you some of the trial-and-error by introducing some important concepts about constructors for those making the jump from Java to Kotlin.

Not All Constructors are Created Equal in Kotlin

In Java, all constructors are equal in a sense.

public class Person {

String name ; int age ;

public Person (Person p) { name = p . name ; age = p . age ; }

public Person (String n , int a) { name = n ; age = a ; } }

Take the above example, Person(Person p) and Person(String name, int age) can be used independently. Of course, you can choose to have one calling another, but it's not required by the language.

In Kotlin there is always a primary constructor. Any additional constructors are secondary constructors. The primary constructor is always incorporated into the class header.

class Person (n : String , a : Int) { var name : String = n var age : Int = a }

The variables name and age are initialized with n and a . In fact, n and a are available anywhere in the class for variable initialization.

class Person (n : String , a : Int) { var name : String = n var age : Int = a var nameX2 : String = n + n

init { println ("The age is: " + a) } }

Any other constructors would become secondary constructors which are required to call the primary constructor in the very beginning.

class Person (n : String , a : Int) { var name : String = n var age : Int = a

constructor (p : Person) : this (p . name , p . age) { } }

The Order of Creation

So, what is the order of processing when calling a secondary constructor? The primary constructor is invoked first, which triggers all the initialization from top to bottom. Then, the body of the secondary constructor is executed.

class Person (n : String , a : Int) { init { println ("1st: initialization block 1 run") }

var name : String = n . apply { println ("2nd: initialization lines run") } var age : Int

init { age = a println ("3rd: initialization block 2 run") }

constructor (p : Person) : this (p . name , p . age) { println ("4th: secondary constructor run") } }

Can I Skip the Primary Constructor?

No, a default primary constructor is still there even when you don't write it and you are still required to call it in the secondary constructor. Also, you cannot initialize the variables in the secondary constructor because it has already passed the initialization timeframe. (Well, they are actually properties, but they behave just like variables in this case.)

class Person () { var name : String /\/ Compile Error: Property must be initialized var age : Int /\/ Compile Error: Property must be initialized

constructor (n : String , a : Int) : this () { name = n age = a } }

OK, if you really want to fake it like a Java constructor, here is the hack. It's NOT recommended and I wrote it just for the learning purpose.

class Person () { var name : String = "" var age : Int = 0

constructor (n : String , a : Int) : this () { name = n age = a } }

What happened above is that we initialized name and age with a default value and assigned them to a new value in the secondary constructor. However, it doesn't work when you replace var with val because Kotlin does not allow variable initialization in a secondary constructor.

Overloading Arguments with Default Value

Last, but not the least. Kotlin has this elegant way to overload arguments with default values.

class Person (n : String = "Nameless" , a : Int = 1) { var name : String = n var age : Int = a }

fun main (args : Array) { var p1 = Person () /\/ name: Nameless, age: 1 var p2 = Person ("Tom") /\/ name: Tom, age: 1 var p3 = Person (a = 5) /\/ name: Nameless, age: 5 }

The primary constructor is actually a sweet requirement in Kotlin. You will always know what is expected to create the object by scanning the the primary constructor without having to look at all the constructors as you would in Java, because constructors are no longer independent of each other. The primary constructor is also easy to spot because it is the first line of the class. If you are starting on learning Kotlin, I think the constructor is a good starting point. I hope you find this post helpful.

Rocket Report: Russia reopens gateway to ISS; Cape Canaveral hosts missile test

Stephen Clark · Ars Technica OpenForum

Welcome to Edition 8.35 of the Rocket Report! The headlines this week are again dominated by the big changes afoot in NASA's exploration program, with the announcement of a Moon base and a nuclear-powered rocket to Mars. The shakeups come as the agency is just a week away from launching Artemis II, a circumlunar flight carrying a crew of four around the Moon. The Ars space team will be writing extensively about this mission in the days ahead, and we may skip the Rocket Report next week to focus on our Artemis II coverage.

As always, we welcome reader submissions . If you don't want to miss an issue, please subscribe using the box below (the form will not appear on AMP-enabled versions of the site). Each report will include information on small-, medium-, and heavy-lift rockets, as well as a quick look ahead at the next three launches on the calendar.

NASA announces nuclear rocket demo. NASA's announcement Tuesday that it will "pause" work on a lunar space station and focus on building a surface base on the Moon was no big surprise to anyone paying attention to the Trump administration's space policy. But what should NASA do with hardware already built for the Gateway outpost? NASA spent close to $4.5 billion on developing a human-tended complex in orbit around the Moon since the Gateway program's official start in 2019. There are pieces of the station undergoing construction and testing in factories scattered around the world. The centerpiece of Gateway, called the Power and Propulsion Element, is closest to being ready for launch. NASA's rejigged exploration roadmap, revealed Tuesday in an all-day event at NASA headquarters in Washington, calls for repurposing the core module for a nuclear-electric propulsion demonstration in deep space, Ars reports .

Read full article

10 films. 10 visionaries. Season 2 of TED Fellows Films is here!

TED Staff · TED Blog

Last year, we launched something new — a fusion of traditional TED Talks paired with short-form documentary. You didn't just watch; you showed up in force. So we're back. Season 2 of the TED Fellows Film Series has officially arrived!

We're thrilled to share a brand-new collection of films spotlighting our 2025 TED Fellows . Meet the pilot training young African women to fly — and rewriting who gets to navigate the skies. Follow the biotechnologist turning organic waste into biodegradable plastics that could reshape our supply chains. Learn from a scientist using satellites to protect crops and fight food insecurity. Step inside a rural American community where another Fellow is rebuilding local economies from the ground up, creating jobs where they vanished decades ago. And that's only the beginning.

These innovators are part of a global network of 500+ TED Fellows across 100+ countries, pushing the boundaries of science, health care, business, journalism, conservation, art, technology and more — whose work touches more than 200 million lives every year.

Season 2 invites you deeper into the Fellows universe: 10 films, 10 powerful stories of people making change possible. The result is electric, human and unmistakably hopeful.

If you want a window into what the future could look like — and who's already building it — start here. Watch all 10 films now at go.ted.com/fellows25 and listen to the extended interviews on the TED Talks Daily podcast here .

> *Each report will include information on small-, medium-, and heavy-lift rockets, as well as a quick look ahead at the next three launches on the calendar.*
>
> Stephen Clark

“Begin PHASE TWO!” – I’m moving to ScienceBlogs…

Ed Yong · Not Exactly Rocket Science (Ed Yong archive)

So, I have some really exciting news – this blog is evolving. After 18 excellent months at WordPress, I am packing up and moving over to ScienceBlogs , a collection of some of the best, er… science blogs on the Interweb.

I want to assure current readers that the blog is not going to change, (well, except in look). Even though the blog will have some academic neighbours, my mission statement of making science interesting and fun to as many people as possible remains the same and the pitch of the writing won’t change.

I still have full freedom to write about whatever I like and if anything, I’m hoping that the scrutiny of a tight community of experienced bloggers, many of whom are hardcore scientists, will push me to ensure an even higher level of accuracy in what I’m putting out.

So for the moment a massive round of thanks to everyone who continues to read and support this blog. The growing traffic and the generally positive comments from people are really gratifying and I’m really excited about the next step.

In a couple of weeks, the new blog should be ready, I’ll post up the new URL, raise my hands in the air, say “Begin Phase Two!” and cackle maniacally.

How chemists turned bourbon waste into supercapacitors

Jennifer Ouellette · Ars Technica OpenForum

Bourbon is a multi-billion-dollar market, but the American barrel-aged whiskey also produces a lot of wasted grain at distilleries. Chemists at the University of Kentucky developed a method to transform that stillage into electrodes and used those electrodes to build supercapacitors with energy storage capacity on par with existing commercial devices. They presented their work at a meeting of the American Chemical Society in Atlanta, Georgia.

US distillers began making bourbon in the 18th century, particularly in Kentucky, but it really took off commercially, in terms of consumption and exports, after World War II. Legally, a whiskey can only be sold as bourbon if its mash is composed of at least 51 percent corn, with any other cereal grain (usually rye and barley) making up the remainder.

The grain is ground up and mixed with water, and mash from a previous distillation is added to create a sour mash. The addition of yeast launches fermentation, after which the mash is distilled to a clear spirit called “white dog.” That spirit is poured into charred new oak barrels for aging of at least two years. It’s the caramelized sugars and vanillin in the charred wood that give bourbon its distinctive dark color and flavor. The barrels are never reused for bourbon, typically being recycled for making barrel-aged beer, wine, and even barbecue and hot sauces.

Read full article

A bit of good news: It's possible to turn around a groundwater crisis

Scott K. Johnson · Ars Technica OpenForum

Generally, when you hear "water use" and "sustainability," you expect those words to be followed by some bad news. Humanity's enduring ability to ignore the math of declining water supplies is almost impressive. But there are cases where actions have successfully reversed our loss of water resources. A new paper in Science by Scott Jasechko of the University of California, Santa Barbara, examines documented cases of groundwater recovery around the world to identify which strategies have worked.

Groundwater is invaluable for many reasons. For one, it's (usually) cleaner than surface water. It's also right under your feet and often close enough to the surface that it doesn't take much energy to pump it up. And there's loads of it down there, no matter the season. Because of this, humans use a lot of it for drinking water, agriculture, and every other use you can think of.

Unfortunately, in many places, the rate of groundwater use has grown to exceed the rate at which precipitation soaks into the ground to replenish it.

Read full article

A unique NASA satellite is falling out of orbit—this team is trying to rescue it

Stephen Clark · Ars Technica OpenForum

BROOMFIELD, Colorado—One of NASA's oldest astronomy missions, the Neil Gehrels Swift Observatory, has been out of action for more than a month as scientists await the arrival of a pioneering robotic rescue mission.

The 21-year-old spacecraft is falling out of orbit, and NASA officials believe it's worth saving—for the right price. Swift is not a flagship astronomy mission like Hubble or Webb, so there's no talk of sending astronauts or spending hundreds of millions of dollars on a rescue expedition. Hubble was upgraded by five space shuttle missions, and billionaire and commercial astronaut Jared Isaacman—now NASA's administrator— proposed a privately funded mission to service Hubble in 2022, but the agency rejected the idea.

Swift may be a more suitable target for a first-of-a-kind commercial rescue mission. It has cost roughly $500 million (adjusted for inflation) to build, launch, and operate, but it is significantly less expensive than Hubble, so the consequences of a botched rescue would be far less severe. Last September, NASA awarded a company named Katalyst Space Technologies a $30 million contract to rapidly build and launch a commercial satellite to stabilize Swift's orbit and extend its mission.

Read full article

Ed vs. Gravity

Ed Yong · Not Exactly Rocket Science (Ed Yong archive)

For the next week, you'll hear tumbleweeds blowing through this blog as I will be on holiday. I'm going to Whistler, Vancouver, where I will be sticking two flimsy strips of wood to my feet and throwing myself down a mountain at high speed. I see it as a challenge to both cold and gravity.

I'll be writing a few things while I'm there so expect some good stuff the week after. For the moment, feel free to scour the Site Index for oldies-but-goodies or have a look at my Review of 2007 for more focused recommendations.

A word about comments: This blog's comments policy are set so that anyone who's had a comment previously approved can post more, but any newbies have to be moderated first. If you've never commented here before and your comment doesn't show up until next week, that's why.

ANALYSIS

IN BRIEF

BBC Sport, BBC Sport: Four-time F1 world champion Max Verstappen tells BBC Sport he is “not enjoying” the sport after changes to the rules this season.

BBC Sport, BBC Sport: Mary Rand, the first British woman to win athletics gold at the Olympics, was described as “Marilyn Monroe on spikes”.

James I. DunlopPeter A. ThomasonLeo M. CarlinBenjamin G. DavisStephen D. CarteraMedical Research Council-University of Glasgow Centre for Virus Research, School of Infection & Immunity, Glasgow G61 1QH, United KingdombCancer Research UK Scotland Institute, Glasgow G61 1BD, United KingdomcSchool of Cancer Sciences, University of Glasgow, Glasgow G12 0ZD, United KingdomdThe Rosalind Franklin Institute, Didcot OX11 0QX, United KingdomeDepartment of Chemistry, Chemistry Research Laboratory, University of Oxford, Oxford OX1 3TA, United Kingdom, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. A defining feature of Rift Valley fever virus (RVFV) is the incorporation of the NSs protein into large filamentous assemblies inside infected nuclei [R. Swanepoel, N. K. Blackburn,J. Gen. Virol.34, 557–561 (1977).], as judged from fixed specimens. To ...

Alexandra LeeAutumn GentzlerDeclan FitzpatrickSithara Raju PonnyOzkan AydemirMyoung Sook HanCaroline A. LewisGuangping GaoRoger J. DavisaProgram in Molecular Medicine, University of Massachusetts Chan Medical School, Worcester, MA 01605bHorae Gene Therapy Center, Department of Genetic and Cellular Medicine, University of Massachusetts Chan Medical School, Worcester, MA 01605, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificanceThe hepatic JUN NH2-terminal kinase (JNK) exhibits mutually exclusive alternative pre-mRNA splicing that results in the expression of JNK spliceoforms. We show that the hepatic adaptive response to the consumption of a high fat diet requires ...

TED Staff, TED Blog: Can’t make it to Atlanta for TEDNext? TED Live brings the minute-by-minute excitement from the TED Theater to your home, school or office. Be one of the first to experience talks from visionary speakers like music mogul Jermaine Dupri, ballerina & activist Misty Copeland, neurosurgeon & reporter Sanjay Gupta, financial educator Vivian Tu and so many more.

Get your TED Live pass to stream the unedited talks in real time starting November 9 , or later on with the on-demand archive.

Jason W. RocksDat P. TruongDmitrij RappoportSamuel Maddrell-ManderDaniel A. Martin-AlarconToni M. LeeSteven CrossanJoshua E. GoldfordaDayhoff Labs, Inc., Cambridge, MA 02140, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificanceWhile nature has evolved enzymes to carry out a vast array of chemical transformations, selecting the ideal protein to initiate an enzyme engineering campaign often presents a significant challenge, slowing progress across biocatalysis and ...

Yingke LiangStephanie A. BuelerJohn L. RubinsteinaMolecular Medicine Program, The Hospital for Sick Children, Toronto, ON M5G 0A4, CanadabDepartment of Biochemistry, The University of Toronto, Toronto, ON M5S 1A8, CanadacDepartment of Medical Biophysics, The University of Toronto, Toronto, ON M5G 2C4, Canada, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificanceMycobacterium tuberculosisand related pathogens possess an unusual cell wall, rich in mycolic acid, that protects them from antibiotics and the host immune system. This study presents the structure of the mycobacterial long-chain acyl-CoA ...

Stephen Clark, Ars Technica OpenForum: Jupiter’s colossal storms generate lightning flashes at least 100 times more powerful than those on Earth, according to scientists analyzing data from NASA’s Juno spacecraft.

Questions about the future of Juno and more than a dozen other robotic science missions began swirling nearly a year ago, when the Trump administration asked mission leaders to submit “closeout” plans for how to turn off their spacecraft. Ars first reported the news soon after the White House released a budget request that called for slashing NASA’s science budget by nearly half.

Read full article

Comments

BBC Sport, BBC Sport: Donegal manager Jim McGuinness is “delighted” as his team claim the county’s first National Football League title in 19 years and second overall with a thumping 3-20 to 2-10 win over Kerry at Croke Park.

BBC Sport, BBC Sport: Scotland continue their World Cup preparations with a friendly against Ivory Coast on Tuesday, but why is this away game being played at Everton’s new stadium in Liverpool?

Vladislav MyrovAlina SuleimanovaSamanta KnapičPaula PartanenMaria VesterinenWenya LiuSatu PalvaJ. Matias PalvaaDepartment of Neuroscience and Biomedical Engineering, Aalto University, Espoo FI-00076, FinlandbNeuroscience Center, Helsinki Institute of Life Science, University of Helsinki, Helsinki FI-00014, FinlandcDivision of Psychology, VISE, Faculty of Education and Psychology, University of Oulu, Oulu G12 8QB, FinlanddBioMag Laboratory, Uusimaa Hospital Diagnostic Center, Helsinki FI-90014, FinlandeSchool of Psychology and Neuroscience, University of Glasgow, Glasgow FI-00029, United Kingdom, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificanceNested neuronal networks exhibit scale-free activity in vivo, but there is a shortage of mechanistic models linking such hierarchical architectures with measurable dynamics. We advance here a Hierarchical Kuramoto model that directly links ...

BBC Sport, BBC Sport: Before Carlos Alcaraz became the youngest man to complete the career Grand Slam, he was simply Carlitos from Murcia. This is his story.

Jing LiuAn MuShaojie MaYi LiuJing ZhangJun-zhuo ZhaoJian WangXia-lian WuJun-xia LuaInterdisciplinary Institute of NMR and Molecular Sciences, Wuhan University of Science and Technology, Wuhan 430081, ChinabSchool of Life Science and Technology, ShanghaiTech University, Shanghai 201210, ChinacSchool of Pharmacy, Jiangsu Ocean University, Lianyungang 222005, ChinadHubei Province for Coal Conversion and New Carbon Materials, School of Chemistry and Chemical Engineering, Wuhan University of Science and Technology, Wuhan 430081, China, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, *As Paul Hoggett eloquently describes, " We are living in a time when a tragedy which is without precedent is unfolding in front of our eyes.*

julietwalker

Northern Ledger

Vol. 1, No. 040 · 2026-03-30

Curated and composed automatically.